Carro Data Insights: July vs. August Forensic Review

The Bottom Line: August's drop in lead-to-inspection conversion (39% down to 31%) is not an ops failure, it's a classic mix-shift trick. Marketing spent more money flooding us with Web leads, which naturally convert lower. However, the profit collapse (Gross Profit crashing from +$3,394 to -$801 per car) is very real. In August, we opened the gates to older, high-mileage cars (2014–2018 models). Every single one of them lost money because workshop refurb costs tripled to over $4,000 while our retail markup shrank to just 6%.

1. Key Findings & Numbers

- Data Cleanup: Dropped 5 duplicate rows at the bottom of transactions, fixed 3 missing buy prices ($0) using initial quotes, and cleaned up 4 negative holding-day glitches without throwing away valid profit data.
- The Conversion Illusion: Total conversion dropped from 38.98% to 30.83%. But Web conversion actually *improved* slightly (23.98% to 25.13%). The drop happened because Web leads surged from 44.8% of our volume in July to 75.9% in August, mathematically dragging down the company average.
- The Real Profit Killer (Older Cars): Cars made in 2018 or earlier went from 21% of our purchases in July to 63% in August. On these older cars, average refurb costs jumped to $4,097, while our front-end markup compressed from $2,771 down to $805. 100% of the 416 older cars bought in August lost money, across every single brand (including Honda and Toyota).

2. What We Need to Do Next

1. Fix Sourcing Rules on Older Cars: Stop buying 2014–2016 models and cars with over 160k km for our retail workshops (send them straight to wholesale instead). For 2017–2018 cars, update our pricing engine to automatically bake in a $3,500 refurb buffer and enforce a minimum 15% retail margin.
2. Rebalance Marketing Spend: Stop blowing the budget on generic Web leads. Shift spend back to WhatsApp and walk-in channels, which convert at nearly double the rate.

3. AI Reflection

- **Tools & Workflow:** Used Gemini to scaffold initial Pandas groupby syntax and formatting templates, while performing data exploration, verification directly in Python.
- **Flawed AI Output & Human Correction:** (1) When asked to clean anomalies, the AI initially wrote a filter (df[df['days_to_sell'] >= 0]) that completely deleted the 4 negative days_to_sell rows in July. Inspecting those rows manually showed their buy, sell, and refurb figures were completely valid, so I changed the logic to set negative days_to_sell to NaN rather than dropping the deals. (2) For margin decay, the AI initially blamed only rising refurb costs on Protons and Peroduas; by checking front-end spread (sell_price - buy_price) across manufacture years myself, I found that retail markup on <=2018 cars had also collapsed from 14.5% to 6.0%, and that 100% of <=2018 cars across all makes (including Honda and Toyota) lost money.